Data Quality Assessment Report

Prepared for Management and Board Review

Metric	Result
Total records	1,283
Original columns	11
Missing values	0
Fully duplicated records	136 records (10.6%)
Age outliers based on IQR	14 records (1.09%)
Age range	45-112 years

Executive Summary

The dataset is complete in terms of actual missing cells: no blank values were detected in the original columns. However, fully duplicated records and statistically high age values were identified and should be validated against the source before final analytical use. The dataset also contains values such as 'Not Specified' and 'Not Recorded'. These are not technically missing cells, but they reduce analytical completeness.

1. Missing Value Check

All 11 original columns were checked and no actual missing values were found. Cell completeness from a null-value perspective is 100%.

2. Duplicate Record Check

A total of 136 excess fully duplicated records were identified across all available columns, representing approximately 10.6% of the dataset. Because the file does not contain a unique identifier, these records should not be automatically deleted. They may represent different cases with identical attributes. The recommended action is to validate them against the source system or an original case identifier, if available outside this file.

3. Outlier Check

The IQR method was applied to the Age column. Q1 = 65, Q3 = 82, and the upper outlier threshold = 107.5 years. Fourteen records have ages from 108 to 112 years. These values are statistically unusual but still logically possible, so they should be verified rather than automatically removed.

Age	Number of records
108	2
109	2

110	6
111	3
112	1

4. Additional Data Quality Observations

• Education level: 382 records are marked as 'Not Specified' (29.8%).

• Admission reasons: 12 records are marked as 'Not Recorded' (0.94%).

• Administrative region: 29 records are marked as 'Not Recorded' (2.26%).

• A spacing inconsistency exists between two values in the education-level category. This may split what should be one category during grouping and reporting.

5. Recommendations Before Final Submission

Validate fully duplicated records against the source before deleting any records. Review ages from 108 to 112 years. Standardize inconsistent categorical text formatting, and review 'Not Specified' and 'Not Recorded' values when alternative source information is available. No new columns were added to the dataset, and the original data was not modified as part of this assessment.

Methodological Note

This report assesses the quality of the uploaded dataset as provided. The visible data relates to care services and older adults; therefore, the assessment is presented as a general data-quality report rather than assuming the file contains sales data.